MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 211W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Mounting position: Any

Weight: 8 ounces (226.8 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	© Ef	Eb	Ec	Ib	Ic	Pp	Pi
Unit:	MHz	Vdc	Vdc	Vdc	mAdc	mAdc	W	W
Maximum:	8	10.5	1,250	-400	175	50	100	220
Minimum:	---	9.5	---	---	---	---	---	---
TEST CONDITIONS:	---	10.0 (Note 4)	1,250	---	---	---	---	---

GENERAL:

© Qualification - Required

© denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	3.1	3.4	Adc
1266	Total grid current	3	Ec/Ib = 100 mAdc; t = 120 sec	0.65	II	Ic	0	-10	µAdc
1256	Electrode current (postive grid)	1	Ef = 9 Vdc; Eb = 250 Vdc; Ec = 100 Vdc	0.65	II	Ic	30	75	mAdc
1256	Electrode current (anode)	-	Eb = 1,000 Vdc; Ec = -50 Vdc	0.65	II	Ib	55	85	mAdc
1236	Power oscillation	-	F = 5 to 15 MHz; t = 120 sec; Ib = 175 mAdc; Rg = 5,000 ohms; Ic = 30 mAdc	0.65	II	Po	135	---	W (useful)
1231	Emission oscillation	3	Power oscillation	0.65	II	Ef	---	9	Vdc
---	Breakdown	2		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
1211	Insulation of electrodes	-	Use 500 volt megger, or equivalent	---	---	R	15	---	Meg
1306	Transconductance	-	Eb = 1,000 Vdc; Ec/Ib = 72 mAdc	---	---	Sm	2,820	4,240	µmhos
1316	Amplification factor	-	Eb = 1,000 Vdc; Ec/Ib = 72 mAdc	---	---	Mu	11	13	---
1236	Internal insulation	-		---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-		---	---	Cgp	12.7	16.3	pF
						Cin	5.0	7.0	pF
						Cout	4.1	6.9	pF
	Quality conformance inspection, part 3								
Ⓒ---	Life test	-	Group C; Eb = 1,000 Vdc (min); Pp = 100 W; Ef = 10 V t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Emission oscillation	-		---	---	Ef	---	9.5	Vdc

NOTES:

1. Secondary emission test; Ec slowly increased from 0 to 100 Vdc; Ic must increase continuously and never fall below any maximum obtained.
2. The tube shall withstand, without breakdown or sparking, the application of 3,000 V effective, at 60 Hz, applied between each pair of elements for 5 seconds.
3. This test is to be the first test performed at the conclusion of the holding period.
Ⓒ 4. The negative filament terminal shall be ground and serve as the reference point for the other dc electrode voltages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

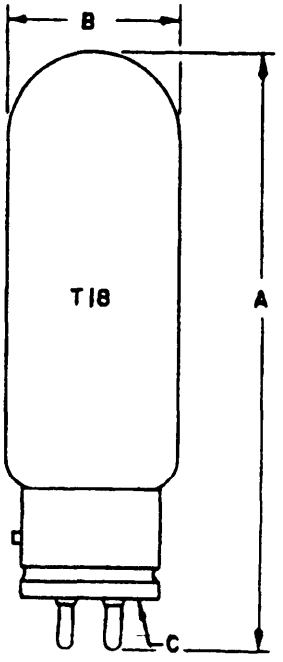
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2945)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	7.875 (200.02)
B	---	2.313 (58.75)
Quality conformance inspection, part 3 (see note)		
C	Base: A4-29 (EIA)	

PIN CONNECTIONS	
Pin No.	Element
1	g
2	f
3	a
4	f

NOTE:
Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

© FIGURE 1. Outline drawing of electron tube type 211W.

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:41Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:41Z
Check the source to verify that this is the current version before use.